Simulation of Feedback Mechanisms Arising From Diversely Seeded Star and Black Hole Formation

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Background

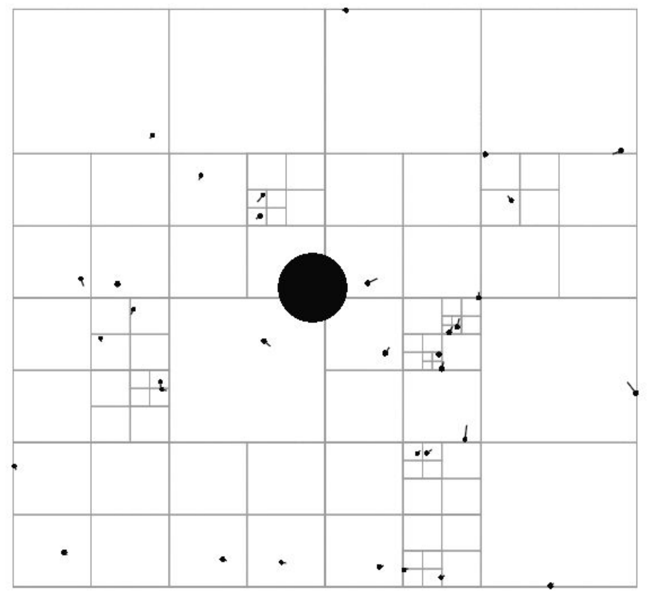
Population III Stars: metal-poor stars in the early Universe

Primordial Black Holes (**PBHs**): black holes from early Universe; very large mass range

- Pop III stars and PBHs are known to influence/feedback each other's formation [1], through enhancing density **fluctuations**, disruptive **tidal forces** and heating, radiative feedback from stars, etc. [2][3]
- Thus must simulate Pop III & PBH evolutions together to capture all effects
- Previous work [2] found effects to vary based on **initial** PBH **mass/abundance**
- Observational constraints on relative PBH abundance (f_{PBH}) can constrain **composition of dark matter** in high-redshift galaxies

Computational Methods

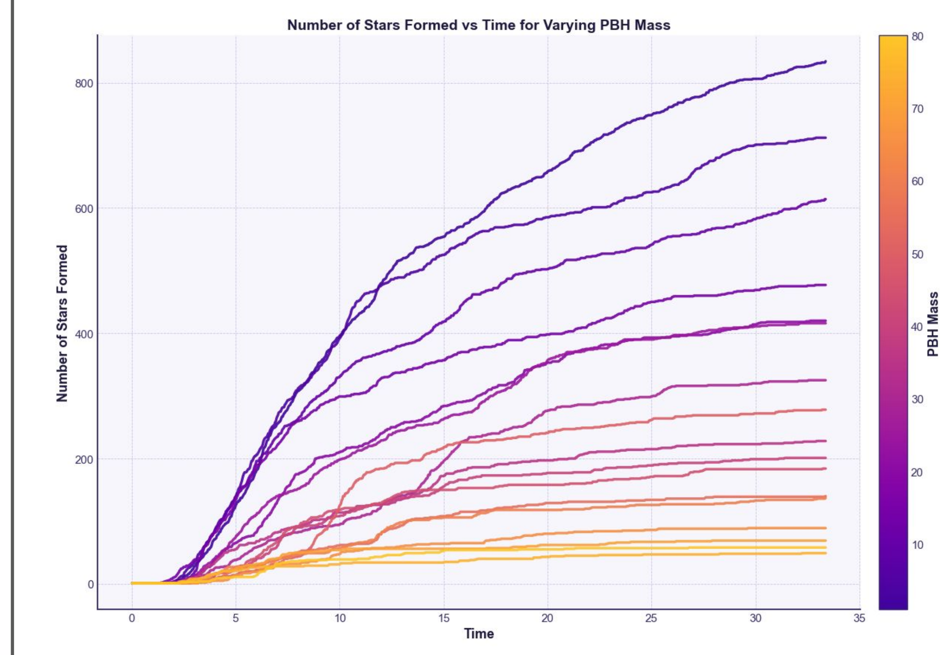
- **Barnes-Hut**-style approximation
  - Approximate long-range forces with center of mass
  - Segment space into **quadtree** (right)
- **Structure of Arrays** approach for memory bandwidth efficiency
- **Parallel task-based** quadtree build and gravity update
- **Cached tree walk** fluid density update
- **Leapfrog** integration (symplectic)
  - Kick-drift-Kick (right)
  - Enables stable orbits by conserving energy

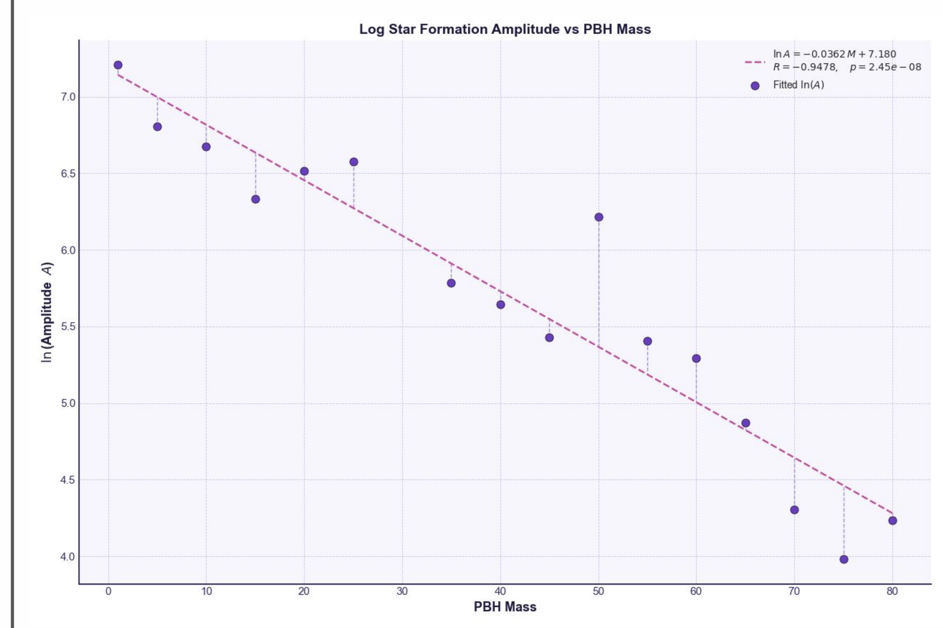


$$\mathbf{v}_{n+1/2} = \mathbf{v}_{n-1/2} + \mathbf{a}_n dt$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_{n+1/2} dt$$

Analysis





- x-axis: **time** after star formation delay expires
- y-axis: number of **stars formed**
- For all PBH masses, star formation is most rapid between t=2 and t=10
- Fitted a **shifted decaying exponential** and extracted A
- ln(A) vs. M plot is **linear** and **decreasing**; total number of stars formed follows decaying **exponential** in M
- Residuals are distributed **randomly** → **good fit** for linear regression

$$N = A(1 - e^{-k(t-t_0)})$$

$$A \sim \exp(aM) \qquad \ln A = aM + b$$

Analytic Methods

Smoothed Particle Hydrodynamics (SPH) simulation [4]

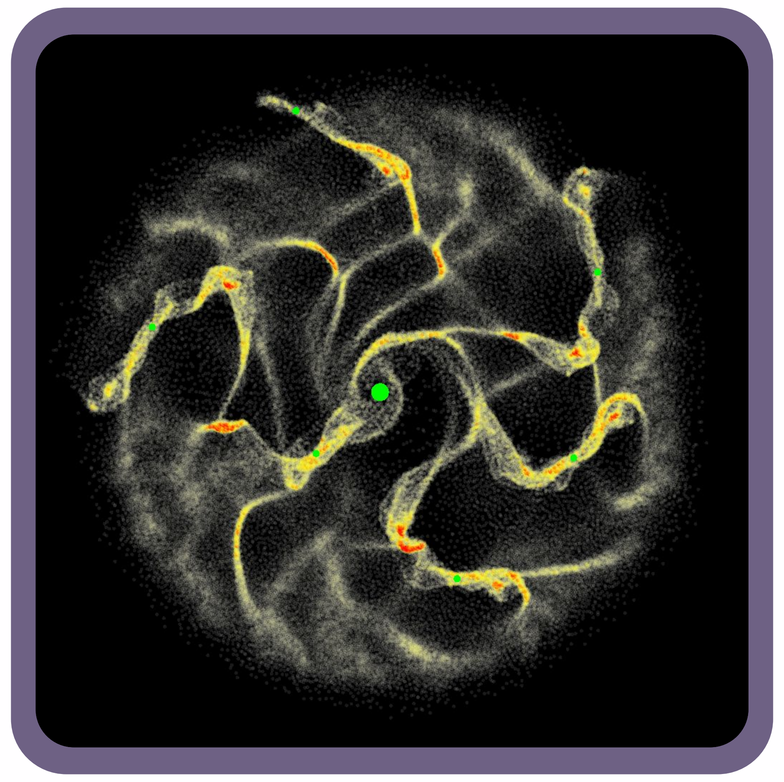
- Nondimensionalized physical quantities
- Store **pressure & density** information with **isothermal** assumption
- Model gas cloud, stars, and PBHs as **particle** masses
- Newtonian Gravitational interactions with softening
- Discretized fluid interaction equations
- **Weighted sum** across nearby particles: Wendland C2 **Kernel** (q: normalized distance)

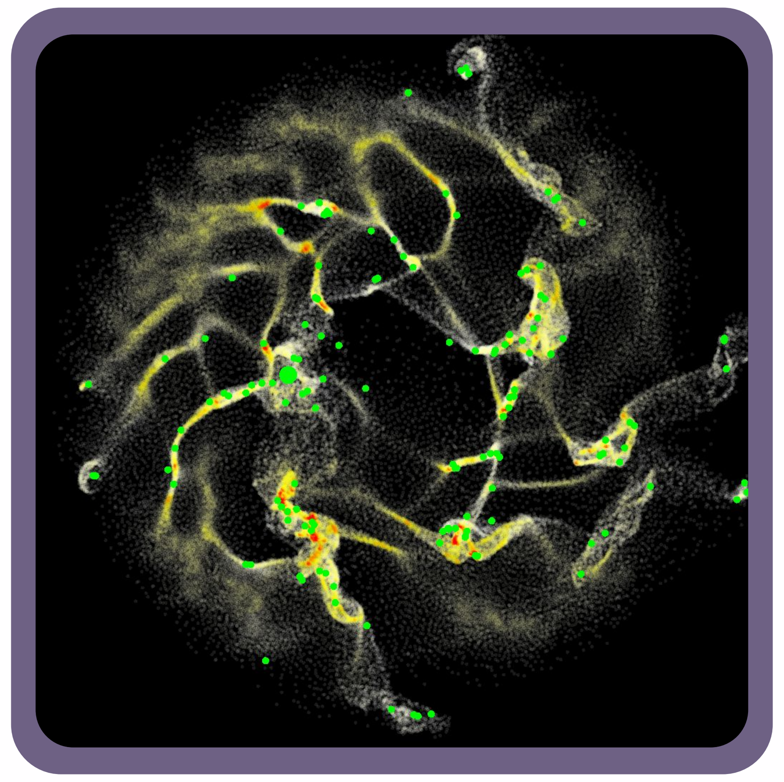
$$W_{ij} = A(1+2q)(1-\frac{q}{2})^4$$

$$\frac{d\mathbf{v}_i}{dt} = \underbrace{-\sum_{j\neq i}^{N} m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2}\right)\nabla W(|\mathbf{r}_i - \mathbf{r}_j|, h)}_{\text{Fluid effects}} + \underbrace{\frac{1}{m_i}\sum_{j\neq i}^{N} F_{ij}\frac{\mathbf{r}_j - \mathbf{r}_i}{\epsilon + r_{ij}}}_{\text{Gravitational effects}}$$

Simulation

m_p	m_{PBH}	N	ρ_J
$5\times10^{-4}\,m$	$0.5\,m$	10^5	$5\pi a(Gh)^{-1}$





Left: earlier time frame and higher central PBH mass. Right: later time frame and lower PBH mass. $m = 10^{34}$ kg; h = kernel width; a = density-to-pressure conversion factor; ρ_J = Jeans critical density.

Conclusions

- **Higher** central PBH masses led to **slower** star formation rate
- Star formation rate **decreased after time** for all mass ranges
- Possible explanations include matter **ejection** by central PBH & higher orbital velocities leading to **greater KE** relative to GPE
- **Future work directions** include more incorporation of the dark matter halo, stellar radiative feedback after formation, and non-monochromatic PBH mass distributions.

Key Citations

[1] Boyuan Liu and Volker Bromm. Accelerating early massive galaxy formation with primordial black holes. *The Astrophysical Journal Letters*, 937(2):L30, September 2022.
[2] Julia Monika Koulen, Stefano Profumo, and Nolan Smyth. Primordial black holes and the first stars. *Physical Review* D, 2025.
[3] Hao Li, Yangyao Chen, Huiyuan Wang, and Houjun Mo. Physical processes behind the co-evolution of haloes, galaxies, and supermassive black holes in the illustrisTNG simulation. *Monthly Notices of the Royal Astronomical Society*, 543(2):1878–1898, September 2025.
[4] J. J. Monaghan. Smoothed particle hydrodynamics. *Annual Review of Astronomy and Astrophysics*, 30:543–574, 1992.